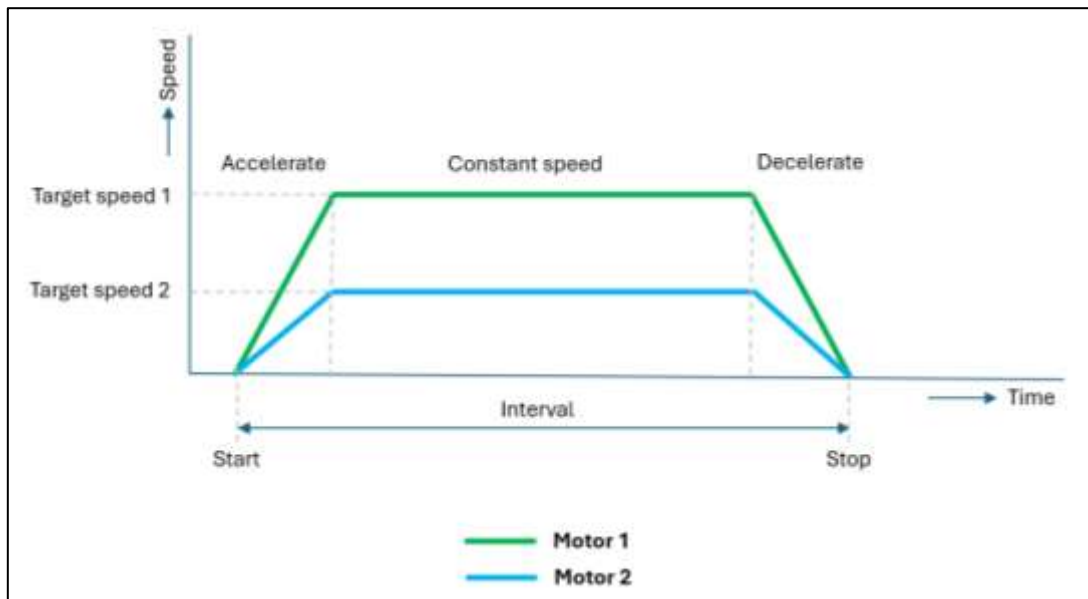


Specification: software for STM32 Nucleo motor controller

1. Introduction

A set of two BLDC motors must be controlled by a STM32 Nucleo-F103RB board. Required cycle see graph below:

- Both motor1 and motor 2 start at the same time and stop at the same time, both with the same given interval.
- Each motor must run at its own given constant speed
- A closed loop control (PID-control) is required to maintain the constant speed per motor at its target speed.



2. BLDC motor specification

- Each BLCD motor has 4 wires
 1. PWM signal with a frequency of 20kHz as an input
 - i. PWM control level “low”: $\leq 0.5V$
 - ii. PWM control level “high” 2.7 ... 5.5V
 2. Tacho signal as an output
 - i. FG-output pulses: 6 pulses / revolution
 - ii. FG-output pulse time “low”: 50%
 - iii. FG-output level “high” (no load): 5 ... 6V
 - iv. FG-output level “low” (no load): $< 0.5V$
 3. + supply voltage (connected to external power supply)
 4. - supply voltage (connected to external power supply)

3. Software requirements for the STM32 Nucleo-F103RB board

- Programmed fixed in the C-program
 - One fixed PWM signal frequency setting of 20kHz for both motor 1 and motor 2
 - Target speed setting of 1900 rpm for motor 1

- Target speed setting of 800 rpm for motor 2
- PID parameter settings for motor 1
- PID parameter settings for motor 2
- One interval setting (time between start and stop) of 35 seconds for both motor 1 and motor 2
- Input 1: Start the interval with an input pulse HIGH (for a momentary push button)
- Input 2: Tacho signal from motor 1
- Input 3: Tacho signal from motor 2
- Output 1: PWM signal for motor 1
- Output 2: PWM signal for motor 2
- Plot speed of motor 1 and speed of motor 2 vs time elapsed using SWV data trace timeline graph in STM32CubeIDE
- Write cumulative rotations of motor 1 and motor 2 after completing a cycle in STM32CubeIDE

4. Required scope of delivery

- Functional software for the STM32 Nucleo-F103RB board as described above
- Software suitable for uploading via STM32CubeIDE version 1.17.0 by us to our own STM32 Nucleo-F103RB board for testing the complete system with the motors and power supply.